

BrickAtlas: A Benchmark for Visual Evidence Following

Supplementary Material

Anonymous CVPR submission

Paper ID Draft

1. Resource and Observation Inventory

This supplement documents the source collection, exact visual observations, assignment audits, qualification protocol and supporting graph controls. It accompanies the current benchmark manuscript. The primary visual dataset remains `brickatlas-display-v3`; subsequent measurement audits do not create a new stimulus version. The 15-model experiment design is a proposal, independent of the sealed earlier three-model study. The separate `complex-examples-v1` records contain the three compositional examples highlighted in the main paper. They have algorithmic GT and source renders but no human or learned-model results, and do not increase the paired visual counts.

The 24 human-authored LDraw sources contain 15,334 placed parts. Source URLs, credits, licenses, original files and hashes are retained. The 617 historical dossiers preserve questions and inspection actions. Only 140 parents contribute primary visual pairs: 73 Color and 67 Part-type. Table 2 lists their source coverage. Table 1 groups the historical resource by assembly scale. The scale bands are not validated cognitive-difficulty levels. Figure 4 presents every source using the original full-assembly renders, with source ID, scale band and part count. Together with the main-paper examples, it separates the physical source collection from its task-specific observations.

Table 1. Source scale and historical item counts. These item counts must not be interpreted as primary paired observations.

Source band	Sources	Parts	Items
D1	5	501	83
D2	9	2,172	175
D3	2	1,440	59
D4	8	11,221	300

1.1. Exact input boundary

An observation binds the complete system and user messages, question, options and native 1280×800 image. The internal record retains its observation ID, source identity, display mapping, condition and gold. The model-visible input does not expose these audit fields. All labels resolve inside the current image. The prescribed output is exactly one JSON object containing one listed `choiceId`. The exact shared identity contract is:

References select objects in the supplied image only. A label such as B0001 is an image-local display reference, not a request to retrieve a historical source instance. Renders may have edited appearances or label assignments. Answer only from the current image.

Changes to text, options, image bytes, dimensions or image policy require renewed qualification of affected observations. Provider envelopes additionally record requested model revision, adapter, generation settings and full request hash. The provider’s internal image preprocessing is not directly observed. A matching local envelope therefore establishes the delivered request, not every internal transformation.

1.2. Concrete visual pair

For OMR 42004, Part-type retains B0035 as target. Candidate B0036 selects the matching source type, 32250, in A. After B0036 and B0116 exchange referents, B0116 selects that same geometry in B. This is a change in the correct reference, not a change in the canonical part’s type.

Figure 1 shows the actual PNGs, verbatim question, complete option lists and GT JSON. Identical crops remove empty margins and retain every operand and label. Table 3 exposes the source-record derivation: only the candidate with part type 32250 matches the target, and its current label determines the choice ID. The source identifier and displayed label must be kept distinct. The GT is generated from records, not asserted to have passed human qualification.

Table 2. All 24 source assemblies. Color and Part-type counts are parents, each contributing exactly one primary pair. Shared parts, authors and derived observations do not increase source count.

Source	Name	Band	Parts	Color	Part-type
31028	Sea Plane	D1	53	2	2
31027	Blue Racer	D1	67	2	2
10156	LEGO Truck	D1	111	2	1
omr-42102	Mini CLAAS XERION	D1	129	2	1
omr-42020	Twin-Rotor Helicopter	D1	141	2	1
10036	Pizza To Go	D2	166	2	2
10014	Caboose	D2	170	2	2
omr-31031	Rainforest Animals	D2	215	2	2
omr-31088	Deep Sea Creatures	D2	230	2	1
omr-42004	Mini Backhoe Loader	D2	246	2	2
omr-42061	Telehandler	D2	261	2	2
31009	Small Cottage	D2	270	2	1
5867	Super Speedster	D2	278	2	2
10128	Train Level Crossing	D2	336	2	2
10159	City Airport	D3	592	3	3
10001	Metroliner	D3	848	4	4
omr-10269	Harley-Davidson Fat Boy	D4	1029	4	4
omr-10242	MINI Cooper	D4	1086	4	4
omr-42066	Air Race Jet	D4	1244	5	5
10213	Shuttle Adventure	D4	1254	5	5
omr-10265	Ford Mustang	D4	1357	5	5
10220	Volkswagen T1 Camper Van	D4	1428	5	4
omr-21309	NASA Apollo Saturn V	D4	1845	5	5
omr-42054	CLAAS XERION 5000 TRAC VC	D4	1978	5	5

Table 3. GT derivation for the supplemental Part-type example. Source instance IDs and part types remain constant; the current label of the matching candidate changes. Options C and D refer to B0036 and B0116 respectively, so the required outputs are {"choiceId": "C"} for A and {"choiceId": "D"} for B.

Source instance	Part type	Label in A	Label in B	Role
B0035	32250	B0035	B0035	Target
B0036	32250	B0036	B0116	Match
B0060	32526	B0060	B0060	Distractor
B0116	32449	B0116	B0036	Distractor
B0196	2780	B0196	B0196	Distractor

visual-evidence.json binds every new figure to source captures or exact observation wires, questions, options, golds and crop rectangles. The archived figure provenance remains available for historical assets. The main paper uses its figure space for the compositional cases; these atomic controls document the frozen visual measurement contract.

1.3. Appearance baseline

The legal Color baseline accepts one wire containing only PNG and text/options. It runs in a fresh isolated process with an empty working directory. The evaluator joins IDs and independently reconstructed gold afterward. No private masks, paths, source facts or counterparts are supplied to

the predictor.

RGB channels are normalized to $[0, 1]$. Near-white pixels with all channels at least .90 are removed. Pixels whose maximum channel is at most .30 vote Black. Other pixels require saturation at least .35 and hue within 60° of the nearest prototype: Red at 0° , Blue at 240° , or Yellow at 60° . Within each category, four-connected components smaller than the greater of 64 pixels and .0005 of image area are removed. The winning category requires .55 of retained support; ties and empty support abstain. Thresholds were locally committed before the corpus run after synthetic tests.

The measured result is 141/146 correct endpoints, five abstentions and 68/73 Both (93.15% micro; 91.875% equal-source). This baseline summarizes rendered appearance

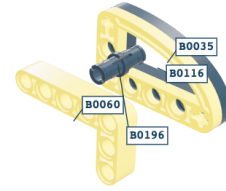
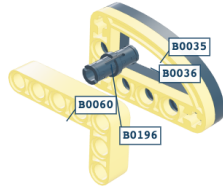
Part-type | Id2-omr-42004-shape-match-1

Which displayed candidate label currently marks an object with the same visible part geometry as the target marked B0035? Ignore color and pose. Answer from this image, not from historical source identities.

A: B0060 B: B0196 C: B0036 D: B0116

Observation A

Observation B



GT: {"choiceId": "C"} = B0036

GT: {"choiceId": "D"} = B0116

GT audit: target B0035 and source instance B0036 share part 32250.
The matching instance is labeled B0036 in A and B0116 in B; all geometry stays fixed.

Figure 1. Atomic Part-type control: fixed target B0035 and all four candidates. The matching source instance is labeled B0036 in A and B0116 in B, changing choice C to D. The common crop retains the remote swapped candidate; no operand is moved or omitted. GT is record-derived, with human visual qualification pending.

across the image; it does not locate the target label or validate multi-object reference following. Per-observation support and failure records are available with the audit.

2. Compositional Examples: Full Inputs and GT

2.1. CX1: minimum restoration of terminal connectivity

Eight vertices are unavailable. Restore the fewest listed vertices so that B0158, B0171, B0179 and B0180 lie in one connected component. A restore action reinstates that vertex and every original edge to a currently present neighbor. Return every minimum repair set, its cost, and the final component sizes. Decide whether budget 2 is sufficient.

The source is the OMR 42004 backhoe subassembly s-629fc43f2db48f6b, containing B0148–B0183. Its induced graph contains 36 vertices and 44 unique undirected recognized connector pairs. Table 4 gives every adjacency before intervention. Paths through the remainder of the vehicle are excluded by the explicit input boundary.

The unavailable set is {B0150, B0160, B0164, B0165, B0167, B0173, B0175, B0177}; terminals are {B0158,

B0171, B0179, B0180}. Each reinstated vertex costs one and recovers every original edge to a present neighbor. The five initial components are listed in Table 5; their sizes are 18, 5, 2, 2 and 1. Contracting these components preserves the restoration problem. The 3D render remains the intact source, avoiding an unsupported physical interpretation of the abstract faults.

GT is the single minimum set {B0160, B0175, B0177}, with cost three. After restoring it, all 31 present vertices lie in one component. We enumerate every subset of the eight candidates: all $2^8 = 256$ outcomes are retained in gold.json. The 37 subsets of cardinality at most two all fail. An independent union-find implementation checks every recorded outcome against the builder’s graph search. This proves minimality under the supplied graph semantics; it does not prove physical disassembly or reassembly.

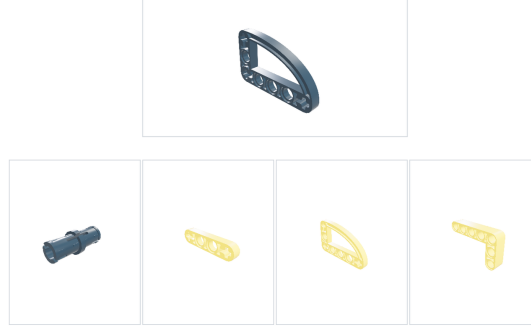
2.2. CX2: conditional inspection with a worst-case budget

Exactly one listed three-vertex fault world holds. A query returns whether its two IDs are connected in the remaining graph. Design a decision tree that identifies the exact world in the minimum worst-case query cost. Specify the first query and both conditional continua-

Position-reference control | Id2-omr-42004-shape-match-1

The upper panel is the target. Which of the four lower candidate panels, read from left to right, shows the same visible part geometry? Ignore color and pose.

A: Leftmost B: Second from left C: Third from left D: Rightmost



GT: {"choiceId":"C"} = Third from left

Figure 2. Auxiliary position-reference question, complete native image, all options and exact GT for OMR 42004. The correct choice is C, third from the left. No labels or GT were added inside the native image. Isolated parts are observations, not a proposed freestanding assembly. The comparison changes layout and visual access together.

Table 4. Complete adjacency input for CX1 and CX2. Edges are undirected, so each recognized pair appears at both endpoints. Every node corresponds to an actual part in the intact source subassembly.

ID	Neighbors	ID	Neighbors
B0148	B0149, B0150, B0151, B0156, B0157	B0166	B0160, B0164, B0167
B0149	B0148, B0155, B0181	B0167	B0163, B0166
B0150	B0148	B0168	B0158, B0163
B0151	B0148, B0152, B0182	B0169	B0160, B0170, B0171, B0172
B0152	B0151, B0153, B0161	B0170	B0169, B0171, B0174
B0153	B0152, B0154, B0155, B0158	B0171	B0169, B0170
B0154	B0153, B0160, B0161	B0172	B0169
B0155	B0149, B0153, B0161	B0173	B0160, B0180
B0156	B0148	B0174	B0170, B0175
B0157	B0148	B0175	B0174, B0176, B0177
B0158	B0153, B0159, B0161, B0162, B0168	B0176	B0175, B0180
B0159	B0158	B0177	B0175, B0179
B0160	B0154, B0166, B0169, B0173	B0178	B0179
B0161	B0152, B0154, B0155, B0158	B0179	B0177, B0178
B0162	B0158, B0163	B0180	B0173, B0176
B0163	B0162, B0164, B0165, B0167, B0168	B0181	B0149
B0164	B0163, B0166	B0182	B0151, B0183
B0165	B0163	B0183	B0182

136 tions. Prove whether a budget of 1 can succeed.

137 CX2 uses CX1’s intact graph. Its worlds are: W1 disables
 138 {B0150, B0160, B0175}; W2 disables {B0150, B0173,
 139 B0175}; W3 disables {B0150, B0164, B0175}; and W4
 140 disables {B0150, B0164, B0165}. Worlds remain fixed
 141 during inspection. Each unit-cost query tests reachability
 142 from B0158: Q1 to B0171, Q2 to B0179, and Q3 to B0180.

Noiseless query outcomes are derived from graph traversal
 (Table 6).

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Table 5. CX1 initial components after all eight specified faults. C1–C5 are diagram labels, not additional parts.

Component	Remaining source IDs
C1	B0148, B0149, B0151, B0152, B0153, B0154, B0155, B0156, B0157 B0158, B0159, B0161, B0162, B0163, B0168, B0181, B0182, B0183
C2	B0169, B0170, B0171, B0172, B0174
C3	B0176, B0180
C4	B0178, B0179
C5	B0166

Table 6. CX2 evaluator outcomes, independently recomputed from the supplied graph and fault worlds. This table is GT, not extra model input.

Query	W1	W2	W3	W4
Q1	no	yes	yes	yes
Q2	no	no	no	yes
Q3	no	no	yes	yes

The optimal first query is Q3. If it returns no, Q1 distinguishes W1 (no) from W2 (yes). If Q3 returns yes, Q2 distinguishes W3 (no) from W4 (yes). Every path costs two. No one-query policy can identify four worlds from a binary result. Checking each possible first query additionally confirms that only Q3 permits a depth-two tree. The GT is evaluated on every compatible world; returning one action sequence is insufficient. The world definitions were selected for branch-dependent diagnosis, not sampled as natural hardware failure frequencies.

2.3. CX3: correspondence, proper registration and probe transfer

Two 12-part axle subassemblies repeat the same CAD design. The four listed anchors have unknown correspondences; IDs provide no matching rule. Match only equal part types. Determine the unique admissible correspondence and its least-squares proper rigid transform $q = R p + t$, with $\det(R)=+1$. A correspondence is admissible only if some proper rigid transform gives maximum anchor error ≤ 0.05 mm. Return the correspondences, R , t , and the mapped origin and local x-axis of B0030. Reflecting the geometry is forbidden.

The two OMR 42061 axle subassemblies each contain twelve placed parts. The four A and four B anchor origins and types appear in the main figure. The input probe B0030 has origin (43.9924, 24.8, 120.4212) mm and local x-axis (−.99938684, 0, −.03501355) in world coordinates. These values are extracted from source transforms, not image measurements, and origins are not bounding-box or mass centers.

Repeated types permit four assignments before geometry is considered. Table 7 reports the proper least-squares fit for each. Only the assignment B0024→B0036,

B0025→B0037, B0026→B0038 and B0035→B0047 is admissible. The least-squares solution is

$$R = \begin{bmatrix} -1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -1 \end{bmatrix}, \quad t = \begin{bmatrix} 16 \\ 0 \\ 120 \end{bmatrix} \text{ mm.}$$

It maps B0030 to (−27.9924, 24.8, −.4212) mm and its local x-axis to (.99938684, 0, .03501355), agreeing with source part B0042.

The anchors are affinely independent: six times their tetrahedron volume is 2046.72 mm³. An independent coordinate-basis solve recovers the same rotation and translation. For every rejected assignment, at least one inter-anchor distance differs by more than .1 mm. The triangle inequality therefore rules out *any* rigid fit within .05 mm per anchor, independent of the optimizer. The accepted fit is verified against all twelve complete source part transforms. The admissible correspondence and exact least-squares solution are unique; the tolerance does not assert uniqueness of every approximate transform.

These three designed examples add explicit composition and optimality requirements. They do not establish calibrated difficulty or a new model ranking. Their human qualification and full study protocol remain future work, separate from display-v3’s fixed denominator.

3. Part-type Assignment and Priors

Admissible configurations enumerate all four-option permutations and nonmatching swaps with an on-screen anchor and unique source-type match. The 4,824 configurations are ordered by parent, option-label tuple and swap. Constraint-equivalent choices keep the minimum integer hash cost, leaving 2,412 variables. One configuration is selected per parent subject to the stated option, transition and shortcut bounds. The recorded single-thread SciPy 1.17.1/HiGHS solution has objective 1,007,668 with zero relative gap. The frozen assignment, rather than runtime-dependent regeneration, defines the released dataset.

Table 7. CX3 candidate assignments and proper least-squares maximum anchor residual. Source IDs are references, not a supplied correspondence rule. Numerical residuals are geometry checks, not model scores.

B-side assignment for (B0024,B0025,B0026,B0035)	Max error (mm)
B0036, B0037, B0038, B0047	0.0000
B0036, B0038, B0037, B0047	5.3315
B0047, B0037, B0038, B0036	9.0207
B0047, B0038, B0037, B0036	8.1726

Table 8. A-to-B choice transitions. Rows are A gold, columns B gold. Left: Color (73 pairs); right: Part-type (67 pairs). Balanced semantic colors do not imply balanced option transitions.

A\B	A	B	C	D	A\B	A	B	C	D
A	0	3	3	7	A	0	6	5	5
B	7	0	7	8	B	6	0	6	5
C	4	8	0	7	C	5	6	0	6
D	5	4	10	0	D	6	5	6	0

Table 9. All declared Part-type oracle-A rules: B accuracy (%). A gold is supplied, so B accuracy equals Both. These are privileged construction audits; their intervals are descriptive after assignment.

Oracle-A rule	Micro	Source	Source 95% CI
First alternative	34.33	26.94	[15.62, 38.61]
Last alternative	32.84	32.01	[19.93, 44.86]
Cyclic next	35.82	38.68	[25.00, 53.33]
Cyclic previous	34.33	28.89	[17.92, 40.35]
Cyclic two	29.85	32.43	[19.79, 46.25]
Smallest alternate label	29.85	24.58	[14.58, 34.38]
Largest alternate label	38.81	39.03	[26.04, 52.50]
Nearest label number	29.85	30.42	[18.75, 43.12]
Farthest label number	31.34	29.03	[17.08, 41.53]
Nearest source center	29.85	36.60	[23.61, 50.97]
Farthest source center	35.82	28.75	[17.92, 40.21]
Nearest projected anchor	25.37	30.14	[17.36, 44.38]
Farthest projected anchor	32.84	27.08	[16.25, 38.75]

Rules use first/last alternative, cyclic option position, alternate label-number extrema, distance from target label number, or nearest/farthest source center or projected anchor. Ties use ascending original numeric labels. Spatial rules use evaluator-side geometry. None reads B pixels, and no rule supplies a universal model ceiling. Independent reconstruction checks golds and features against source records and rendering metadata.

4. Human Qualification

4.1. Assignment and raw records

Six packages contain 115, 116, 114, 115, 115 and 119 judgments. Their union covers all 347 observations exactly twice. Reviewers must be distinct people, and each sees at most one observation of any parent, including the panel control. Three-observation parents therefore use all six reviewers. Queues are shuffled and use opaque unit IDs without condition or arm labels.

A raw record contains the reviewer pseudonym, timestamp, observation hash, visual choice, categorical decision, four structured checks, reason, native-size inspection attestation and independence attestation. The four checks concern a unique target, unique visual answer, sufficient visibility and absence of reference contradictions. Decidable judgments require a listed choice and all checks passing. Other decisions require an exclusion reason. The interface hides gold, source data, counterpart, model output and other reviewers' judgments. The coordinator retains unmodified exports.

4.2. Adjudication and denominators

Agreement on decision, choice and checks permits finalization. Any disagreement on these fields requires an additional independent adjudicator. Up to three adjudicator slots preserve parent-disjoint exposure and exclude initial reviewers. The adjudicator sees the single observation and its two judgments, but not gold, another observation of that parent or model outputs. Original records remain intact. A final decidable choice that conflicts with generated gold blocks finalization and requires stimulus revision and renewed review.

The full primary universe is always 280 endpoints and 140 pairs. There are 560 initial primary judgments and 134 auxiliary judgments. Actual collected judgments are currently zero. The runner's existing full-study gate requires final decisions for every planned endpoint. Nondecidable cases remain in full-set scoring; the additional qualified subset requires two decidable endpoints per pair. A new MVP needs a separate frozen subset with valid review lineage, not an override of the full-study gate.

Raw categorical agreement and choice/check agreement

use their actual eligible denominators. Pooled two-slot Cohen’s kappa is $(p_o - p_e)/(1 - p_e)$ from empirical decision marginals. Because assigned people vary across observations, it is not the reliability of one fixed reviewer pair. Missing judgments or degenerate expected agreement produce undefined kappa. Adjudicated agreement cannot replace the raw measure.

5. Scoring and Dependence

The four B outcomes partition every changed-answer pair: correct new gold, valid old gold, another valid answer, and invalid output. Missing calls are invalid and remain in the fixed denominator. AAcc and Both retain information about A correctness. Conditional adaptation reports the number of A-correct pairs and the sources with at least one eligible pair.

Equal-source estimates average within-source means. The bootstrap resamples whole sources 10,000 times using seed 20260920. Model contrasts use the same pairs. Qualified-minus-full contrasts reuse source weights and recompute means on eligible sources. An empty subset or a draw containing no eligible source is recorded as undefined. No whole-source interval estimates provider randomness across repeats. Boundary outcomes can yield degenerate intervals.

The 24 sources form 18 primary-author and 17 contributor-connected groups. Contributor groups include credited OMR editors. Sensitivity analysis averages within-source outcomes by author, resamples whole author groups and omits each author in turn. Direction changes under omission limit stability claims. The public inventory additionally contains all 276 source-pair part-count and part-set Jaccards; these are inventory similarities, not spatially aligned geometry overlaps.

Table 10. Repeated primary authors. Other primary-author groups contain one source each.

Repeated primary author	Source IDs
Merlijn Wissink	31027, 31028
Philippe Hurbain	42054, 42061, 42066
Robert Paciorek	10036, 10128, 10156
Zoltan Keri	10001, 10159

Table 11. Largest part-inventory overlaps. Count Jaccard uses summed per-type minima divided by summed maxima; set Jaccard ignores counts.

Source A	Source B	Count J.	Set J.
42054	42066	0.450	0.511
42004	42061	0.370	0.354
42004	42020	0.325	0.431
10220	10242	0.322	0.257
10265	10269	0.280	0.320

6. Graph Construction and Completed Algorithm Audits

Graph questions provide explicit undirected edge lists and a node to delete. Repeated edges are deduplicated; surviving isolates count as components. Independent BFS and union-find agree on all golds. These checks validate the supplied graph operation, not completeness of physical connector detection.

The hypothetical collection has 219 distinct observations: an anchor, a changed partner and an isomorphic invariant partner for each of 73 parents. Strong change contains 12 pairs from eight sources; degree-visible change contains 61 from 23 sources. Each has a corresponding invariance partition. Pair endpoints therefore have 292 appearances across all partitions, but only 219 unique inputs.

Table 13. All strong changed-answer pairs, with matching surviving nodes, edge count, target degree, degree multiset and isolate count. BFS and union-find have Both=1 on each.

Parent task	y_A	y_B	Isolates+1 Both
ld2-10001-graph-removal-4	2	3	0
ld2-10014-graph-removal-1	2	3	0
ld2-10014-graph-removal-2	1	2	0
ld2-10213-graph-removal-1	4	3	0
ld2-31009-graph-removal-2	4	3	0
ld2-5867-graph-removal-1	1	2	0
ld2-omr-10269-graph-removal-1	2	1	0
ld2-omr-10269-graph-removal-2	2	1	0
ld2-omr-10269-graph-removal-4	2	1	0
ld2-omr-21309-graph-removal-2	2	3	0
ld2-omr-21309-graph-removal-4	2	1	0
ld2-omr-42054-graph-removal-4	3	4	0

The strong construction compares a six-cycle and two triangles plus identical isolated nodes. The weaker construction compares a path and a triangle plus isolates. Only the former matches the complete degree multiset. Uniform-feature anonymous 1-WL signatures collide on every strong pair, so a deterministic signature-only predictor cannot answer both different golds correctly. Unique node identifiers,

Text-graph task | Id2-10014-graph-removal-2

Delete B0024 and its incident edges from this local graph. How many connected components remain? Do not infer physical extractability.



Matched on both sides: 6 surviving nodes, 6 edges, degree multiset [2, 2, 2, 2, 2, 2], no isolates.

Figure 3. Small matched graph control from the frozen graph track: one six-cycle versus two triangles after deleting B0024. The supplied text edge lists determine GT. CX1–CX2 in the main paper use the larger 36-vertex source graph for optimization and conditional diagnosis; they are a separate set of examples.

Table 12. Completed deterministic algorithm audit. These values are computed algorithm outputs, not planned or measured language-model results. Natural and hypothetical lookups use separate source-held-out fits. Invariance shares anchors with change.

Algorithm	Natural correct / 73	Strong change Both / 12	Degree-visible Both / 61	Strong invariance Both / 12
Node-count lookup	63	0	0	6
Edge-count lookup	44	0	0	7
Degree-histogram lookup	69	0	61	6
Anonymous 1-WL lookup	69	0	61	6
Isolates + one	72	0	61	6
BFS	73	12	61	12
Union-find	73	12	61	12

positional features, spectral features and higher-order models lie outside this restriction.

Feature lookups train only on other sources, choose the most frequent answer, break ties toward the smaller integer and abstain on unseen features. Abstention is incorrect. Hypothetical fitting includes the changed endpoints once and excludes invariant duplicates. The isolates-plus-one heuristic predicts the isolate count plus one when any nonisolated node remains; it never traverses components.

Table 14. Completed invariance algorithm audit. Repeating a wrong answer is consistent but fails Both.

Algorithm	Strong / 12	Degree-visible / 61
Node-count lookup	6	31
Edge-count lookup	7	32
Degree-histogram lookup	6	61
Anonymous 1-WL lookup	6	61
Isolates + one	6	61
BFS	12	61
Union-find	12	61

Strong change has six increases and six decreases in gold component count; arm-wise answer histograms need not match. Both is invariant to swapping A and B. Wilson intervals for 0/12 and 12/12 are approximately [0, .2425] and [.7575, 1] under a pair-independence approximation that source reuse violates. The single template and eight-source coverage limit generalization.

7. Collection and Publication Contract

The main paper lists 15 proposed models and 306 empty result cells across primary visual, diagnostic, control, graph and mode tables. The model inventory comes from the existing budget snapshot. All result values in `experiments.json` are null and render as empty cells. Null represents uncollected results, not measured zero or an unspecified model name.

Before collection, the new study must freeze selected IDs, exact models and revisions, provider/precision, adapter code, image policy, output and effort settings, repeat count, timeouts, inclusion and failure policies, and qualification lineage. The MVP proposal keeps complete visual pairs and all three observations of each selected graph parent. It requires new adapters and a subset qualification gate; those implementation steps remain open.

The earlier sealed evaluator fixes three models and nine runs. It cannot silently accept the new roster. Its receipt reconstruction, failure denominators and source estimators provide implementation references, while the new study requires its own versioned manifest. No synthetic fixture, dry run, historical response or missing receipt can substitute for a genuine observation in empirical tables. Real response cases must link to exact requests and retained outputs; an absent failure category remains absent.

Current manuscript files are under `paper/`; code and dataset indexes are under `benchmark/`. Intermediate reviews, guidance and manuscripts are physically archived under `tem/`, with compatibility links for historical paths. Git includes data, image assets, reviewer packages, code and PDFs. Redundant local recovery tarballs and runtimes are excluded; full historical recovery checks require those separate bundles. Hashes establish local consistency, while genuine human independence and provider origin additionally require collection oversight.

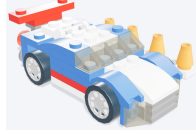
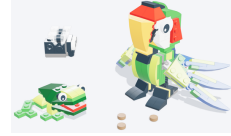
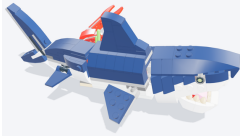
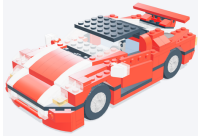
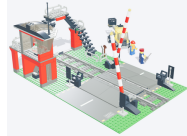
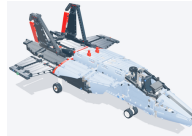
D1 | 31028 | 53 parts
Sea Plane**D1 | 31027 | 67 parts**
Blue Racer**D1 | 10156 | 111 parts**
LEGO Truck**D1 | 42102 | 129 parts**
Mini CLAAS XERION**D1 | 42020 | 141 parts**
Twin-Rotor Helicopter**D2 | 10036 | 166 parts**
Pizza To Go**D2 | 10014 | 170 parts**
Caboose**D2 | 31031 | 215 parts**
Rainforest Animals**D2 | 31088 | 230 parts**
Deep Sea Creatures**D2 | 42004 | 246 parts**
Mini Backhoe Loader**D2 | 42061 | 261 parts**
Telehandler**D2 | 31009 | 270 parts**
Small Cottage**D2 | 5867 | 278 parts**
Super Speedster**D2 | 10128 | 336 parts**
Train Level Crossing**D3 | 10159 | 592 parts**
City Airport**D3 | 10001 | 848 parts**
Metroliner**D4 | 10269 | 1,029 parts**
Harley-Davidson Fat Boy**D4 | 10242 | 1,086 parts**
MINI Cooper**D4 | 42066 | 1,244 parts**
Air Race Jet**D4 | 10213 | 1,254 parts**
Shuttle Adventure**D4 | 10265 | 1,357 parts**
Ford Mustang**D4 | 10220 | 1,428 parts**
Volkswagen T1 Camper Van**D4 | 21309 | 1,845 parts**
NASA Apollo Saturn V**D4 | 42054 | 1,978 parts**
CLAAS XERION 5000 TRAC VC

Figure 4. All 24 source assemblies, retaining original LDraw geometry and saved poses. IDs and counts are read from the source catalog; the images are hash-verified original browser captures. Scale bands describe source size. These views are for resource inspection and are not substituted for frozen primary model inputs.